

The Naylor Model

Variable Glossary

Statcast-style reference · v7

Every metric here earns its place. For each one:

- what it measures (plain English)
- units, league average, and what's elite
 - a real player example
- why it moves steal success

Speed & Acceleration

Sprint Speed

Top running speed measured during the player's fastest sprints.

Units: feet per second (ft/s) · **League avg: 27.0** · **Elite: 29.0 +**

Tiers: Elite 29.0 + | Above avg 27.5 – 29.0 | Average 26.5 – 27.5 | Below avg 25.5 – 26.5 | Poor $\leq$ 25.5

e.g. Bobby Witt Jr.: 30.2 (elite) · Josh Naylor: 24.4 (3rd %ile)

Faster runners are obviously harder to throw out, but the model finds it matters LESS than raw acceleration on a 90-ft steal. Once you're above ~28 ft/s the extra speed is wasted because you're past second base before the throw arrives anyway.

Speed Capped

Sprint speed, but anything over 28 ft/s is clipped to 28.

Units: ft/s · **League avg: 27.0** · **Elite: 28.0 (capped)**

Tiers: $\geq$ 28 ft/s treated as 28.0 (no extra reward) | $<$ 28 ft/s kept as-is — every fraction matters

e.g. Witt 30.2 → capped at 28.0 · Naylor 24.4 → unchanged

The model adds this version because the marginal benefit of more speed vanishes above 28 ft/s. Capping prevents the model from over-rewarding pure speedsters and lets it find smaller advantages like jump and lead.

Jump Time (accel_0_30)

How fast you cover the first 30 feet from contact.

Units: seconds · **League avg: 1.78** · **Elite: $\leq$ 1.65**

Tiers: Elite $\leq$ 1.68 | Above avg 1.69 – 1.74 | Average 1.75 – 1.80 | Below avg 1.81 – 1.86 | Poor $\geq$ 1.87

e.g. Corbin Carroll: 1.67 · Josh Naylor: 1.87 (slow) · Nasim Núñez: 1.64 (elite)

The single most important running metric for steals — the throw is in the air before you hit top speed, so what happens in the first 30 ft decides everything. In the v5/v6 simple GLM this feature has the strongest weight of any speed variable.

Accel Gap

How much faster the runner is off the line than their top speed implies.

Units: percentile points · **League avg: 0** · **Elite: + 25 or more**

Tiers: Naylor archetype +15 or more (fast jump despite low speed) | Above expected +5 to +15 | As expected –5 to +5 |

Below expected –15 to –5 | Pure speedster $\leq$ –15 (fast top speed but slow start)

e.g. Naylor: +22 to +30 across seasons · Witt: –20 (slower off the line for his speed)

This is the heart of the Naylor archetype. POSITIVE values mean the runner is faster off the line than their top speed would predict — exactly what a slow-but-effective stealer looks like. Pure speedsters often have NEGATIVE accel_gap because they coast on top-end speed.

Accel→Top-Speed Premium (v7)

How few feet a runner needs to reach top speed — rewarded MORE when that top speed is genuinely fast.

Units: speed-adjusted index (higher = better) · **League avg: 0** · **Elite: + 5 or more**

Tiers: Premium reaches top speed early FOR a fast runner | Above expected earlier than speed-peers | As expected ~ league runway for that speed | Below expected needs more runway than peers | Long wind-up slow to reach top speed
e.g. Naylor 2025: reaches near-peak in ~50 ft — sooner than his speed predicts (premium +7.7) · Freeman 2024: ~65 ft (negative premium)

Two ingredients: (1) `dist_to_top_speed_ft` — the distance at which a runner first hits 97% of their realized top velocity; (2) a speed-expected baseline (quadratic in sprint speed). The gap = actual – expected; reaching top speed sooner than peers is good. Because high top speeds need more runway, a small gap at high speed is rare and valuable, so the premium scales up with sprint speed. Added to the GBM, the GLM weight table, and as a 9th SSSI feature in v7.

Distance to Top Speed (`dist_to_top_speed_ft`, v7)

Runway (in feet) before the runner first reaches 97% of their own top sprint velocity.

Units: feet (ft) · **League avg: ~65** · **Elite: ≤ 55 (reaches near-peak on less runway)**

Tiers: Very early ≤ 50 ft | Early 51 – 60 ft | Average 61 – 67 ft | Late 68 – 75 ft | Very late ≥ 76 ft
e.g. League runs from ~40 ft (earliest) to 90 ft; most cluster 60–70 ft

The raw input behind the Accel→Top-Speed Premium. Measured as the distance at which a runner first hits 97% of their OWN in-run peak velocity — most MLB runners are still building speed through the home-to-first run, so this clusters in the 60s. What matters for the model is the speed-adjusted RESIDUAL, not the absolute number.

Lead Distance (real Baseball Savant)

Primary Lead

Average distance off first base when the pitcher starts his motion.

Units: feet (ft) · **League avg: 11.5** · **Elite: 13.0 +**

Tiers: Aggressive 13.0 + | Above avg 12.0 – 13.0 | Average 11.0 – 12.0 | Conservative 10.0 – 11.0 | Very small ≤ 10.0
e.g. Trea Turner: 13.1 (elite) · Mookie Betts: 12.4 · Naylor: 9.8 (small lead)

Bigger lead = less distance to cover. But coaches push leads to where the pitcher gets nervous, so big leads can also increase pickoff risk. Notably Naylor has one of the SMALLEST primary leads in MLB (9.8 ft) yet steals 90%+ — his edge is elsewhere.

Secondary Lead

Average distance off first base at the moment of pitch release.

Units: feet (ft) · **League avg: 14.5** · **Elite: 16.0 +**

Tiers: Elite 16.0 + | Above avg 15.0 – 16.0 | Average 14.0 – 15.0 | Below avg 13.0 – 14.0 | Poor ≤ 13.0
e.g. Mookie Betts: 18.1 (elite) · Naylor: 13.7

Primary lead PLUS ground covered while the pitcher is delivering. Big secondary leads compress the steal distance AND give the runner momentum.

Lead Gain (the 'jerk' metric)

Distance covered between pitcher's first move and pitch release.

Units: feet (ft) · **League avg: 3.5** · **Elite: 4.5 +**

Tiers: Elite 4.5 + | Above avg 3.8 – 4.5 | Average 3.2 – 3.8 | Below avg 2.6 – 3.2 | Poor ≤ 2.6

e.g. *Mookie Betts: 5.6 (elite — reads pitchers)* · *Naylor: 3.9 (above avg)*

The user's 'jerk' metric. Captures TWO things at once: how well the runner READS the pitcher's first move AND how explosively they take their secondary lead. Big lead-gain is a hallmark of smart baserunners (Betts, Chisholm Jr., Frazier, Arraez).

v5 / v6 NEW Metrics

Pre-Release Velocity (pre_rel_vel)

How much ground the runner covers per second between pitcher first move and pitch release.

Units: ft / sec · **League avg: 2.7** · **Elite: 3.5 +**

Tiers: Elite 3.5 + | Above avg 3.0 – 3.5 | Average 2.5 – 3.0 | Below avg 2.0 – 2.5 | Poor ≤ 2.0

e.g. *Jazz Chisholm Jr. 2025: 4.34 (elite)* · *Naylor: 2.98* · *Soto: 2.39*

Even if a pitcher has a fast delivery, a runner who covers a lot of ground in that small window is still dangerous. We use a league-constant delivery (1.30 s) as the divisor because per-pitch pitcher TTP isn't publicly available — so variation is driven by Lead Gain. The leaderboard separates 'smart' baserunners from pure speedsters.

Post-Release Distance (post_rel_dist)

How much ground the runner covers during the catcher's pop time.

Units: feet (ft) · **League avg: 51.5** · **Elite: 58.0 +**

Tiers: Elite 58.0 + | Above avg 54.0 – 58.0 | Average 50.0 – 54.0 | Below avg 46.0 – 50.0 | Poor ≤ 46.0

e.g. *Victor Scott II 2025: 59.5 (elite)* · *Naylor: 41.5 (last %ile)*

After the pitcher releases the ball, the runner has ~1.95 seconds (the catcher's pop time) to cover as much ground as possible before the throw arrives. Slower runners still accelerating during this window cover LESS than sprint $\times$ pop implies — we subtract an acceleration tax for them. This is the speedster-archetype metric: Witt, Carroll, Scott II dominate.

Real Stolen-Base Performance

SB / CS

Hard counts: stolen bases and caught stealings in the season.

Units: count · **League avg: varies** · **Elite: 30 SB +**

Tiers: Elite volume 30 + SB | Above avg 20 – 30 SB | Average 10 – 20 SB | Low < 10 SB (excluded from model)

e.g. *Naylor 2025: 30 SB / 2 CS = 93.8% success*

Ground truth. We require $SB + CS \geq 10$ in a season for the runner-season to qualify (smaller samples are too noisy).

Real SB % (shrunk)

Stolen-base success rate, adjusted to prevent tiny samples from looking 100%.

Units: fraction (0-1) · **League avg: 0.78** · **Elite: 0.85 +**

Tiers: Elite 0.88 + | Above avg 0.80 – 0.88 | Average 0.72 – 0.80 | Below avg 0.65 – 0.72 | Poor $\leq$ 0.65
e.g. Naylor 2025: 0.917 (30 SB, 2 CS)

Shrinkage: a 1-for-1 runner doesn't look like 100%. Formula: $(SB + 5 \cdot \text{league\%}) \div (SB + CS + 5)$. League-mean prior 'pulls' small samples toward 78%.

Expected SB %

The SB% you'd expect from this runner based ONLY on their speed.

Units: fraction (0-1) · **League avg: $\approx$ Real SB %** · **Elite: n/a**

Tiers: Used as a baseline compare against Real SB% to get residual
e.g. Naylor 2025 expected: 0.79 (3rd %ile speed) · actual: 0.92

Baseline. Subtracting from Real SB% gives the 'speed-adjusted skill' signal.

SB Residual (the KEY signal)

How much better (or worse) the runner is than their speed predicts.

Units: fraction (typically -0.20 to $+0.20$) · **League avg: 0** · **Elite: $+0.10$ or more**

Tiers: Massive overperformer $+0.10$ + | Above expectation $+0.04$ to $+0.10$ | As expected -0.04 to $+0.04$ | Below expectation -0.10 to -0.04 | Far below ≤ -0.10
e.g. Naylor 2025: $+0.122$ (huge overperformer) · Soto 2025: $+0.128$

POSITIVE values mean the runner steals MORE successfully than their raw speed suggests. This is the Naylor / Soto story. We weight it heavily in the SSSI index.

Expected SB Outcome (xSB · v7)

Expected SB Outcome (xsb_outcome, v7)

Combined speed-and-production score: are you BOTH fast AND a productive base-stealer?

Units: sum of two z-scores (≈ -4 to $+4$) · **League avg: 0** · **Elite: $+2.5$ or more**

Tiers: Realized Burner $+2.5$ + (fast AND productive) | Strong $+1.0$ to $+2.5$ | Average -1.0 to $+1.0$ | Weak -2.5 to -1.0 | Stationary ≤ -2.5
e.g. Elly De La Cruz: top of the board (elite speed + huge volume) · Naylor: positive on production, negative on speed $\rightarrow$ modest xSB

A COMPLEMENTARY lens to the SSSI. Where the SSSI surfaces slow-but-skilled stealers (the Naylor archetype), xSB surfaces the high-ceiling runners who combine raw wheels with real production. It is descriptive, not predictive — kept out of the GBM (z(SB) would leak the outcome) and the SSSI.

SB Potential Gap (sb_potential_gap, v7)

Fast but under-stealing? The gap flags untapped base-stealing potential — coachable upside.

Units: difference of two z-scores · **League avg: 0** · **Elite: n/a — read by sign, not magnitude**

Tiers: Untapped Wheels large + : fast, under-steals | Some upside small + | Balanced ≈ 0 | Over-performs small - |

Crafty Technician large - : steals despite modest speed

e.g. Big POSITIVE = elite speed, few steals (unleash them) · NEGATIVE = over-performs speed (Naylor / Soto archetype)

The analytic payoff of xSB. Positive values are the clearest coaching targets in the model — the tools are already there, the production isn't yet. Negative values are the Naylor quadrant: production that outruns the radar gun.

Battery Context (real per-attempt)

Catcher Pop Time (pop_2b_sba)

Catcher's time from glove to second base on a steal attempt.

Units: seconds · **League avg: 1.95** · **Elite: ≤ 1.85**

Tiers: Elite arm ≤ 1.85 | Above avg 1.86 – 1.92 | Average 1.93 – 1.97 | Below avg 1.98 – 2.04 | Poor ≥ 2.05

e.g. Patrick Bailey: 1.85 (elite) · league avg: 1.95

REAL per-catcher-per-year data starting 2018. Low pop time = throw arrives at 2B sooner = harder to steal. In Model A this is one of the top 5 features.

Composite Scores

SSSI v7 (Slow-Steal Skill Index)

A weighted composite score that identifies slow-but-effective stealers.

Units: z-score units (typically -3 to +3) · **League avg: 0** · **Elite: +1.8 or more**

Tiers: Elite slow-steal +1.8 + | Above avg +1.0 to +1.8 | Average -0.5 to +1.0 | Below avg -1.5 to -0.5 | Poor ≤ -1.5

e.g. Naylor 2025: +1.85 · Soto 2025: +1.18 · league mean: 0

Weighted average of nine standardized features (v7 adds the Accel→Top-Speed Premium). Designed to find the Naylor / Soto archetype: elite-performing slow runners. Weights were tuned on 80% of runners with Naylor + Soto HELD OUT to avoid overfitting.

How to Read the Model's Weight Table

SB % Boost per Tier (standardised logit coefficient)

If a runner improves this feature by 1 tier (1 standard deviation), how much does the model's predicted SB success rate go up?

Units: percentage points (pp) · **League avg: 0** · **Elite: +5 pp or more = a strongly helpful feature**

Tiers: Strongly helps +5 pp + | Helps a little +1 to +5 pp | Neutral -1 to +1 pp | Hurts a little -5 to -1 pp | Hurts a lot ≤ -5 pp

e.g. Jump Time has -10.6 pp → improving Jump Time by 1 tier raises predicted SB% from 62% to 51% (-10.6 pp). Wait, that's WORSE? No — Jump Time is in seconds, and LOWER is better. So a 1-tier INCREASE (slower) hurts by 10.6 pp. The model shows direction with a sign.

Replaces the old 'coef_z' column. Plain-English version of the same number. Compute as:

$P_{\text{baseline}} = \text{sigmoid}(\text{intercept})$

$P_{\text{after}} = \text{sigmoid}(\text{intercept} + \text{coef})$

$\text{boost_pp} = (P_{\text{after}} - P_{\text{baseline}}) \times 100$

Understanding the Results

The SB Residual: The Core Signal

Why Not Just Use Success Rate?

A runner with a 90% success rate is not automatically better than one with 80%. If the first runner only steals against catchers with slow pop times and pitchers who ignore runners, the 90% might be expected given context. The 80% runner might be attempting far tougher steals. The model's primary signal — called the SB Residual — corrects for this by asking: given this runner's sprint speed, what success rate would we predict for an average runner at that speed, and how much does this specific runner beat or miss that expectation? The residual is simply (real SB%) minus (speed-expected SB%). Positive values mean the runner outperforms their speed peers. Naylor's residual is large and positive. Trea Turner's is smaller than you might expect because his success rate is exactly in line with what a 30 ft/s runner should be able to do.

Bayesian Shrinkage: Why Small Sample Rates Are Distrusted

A runner who goes 1-for-1 has a 100% success rate, but that number is meaningless. The model applies Bayesian shrinkage with a constant of $k=5$: every runner's success rate is computed as $(SB + 5 \times \text{league_average}) / (SB + CS + 5)$. This pulls small-sample rates toward the league mean and only lets extreme rates stand when the sample is large enough. A runner with 30 attempts and 28 successes (93.3%) stays near 93.3% because the sample is real. A runner with 1 attempt and 1 success (100%) gets pulled back toward 78%.

The Three Models — A, B, and C

Why Are There Multiple Models?

The pipeline produces three separate models for different purposes. They are not three versions of the same thing; each answers a different question, and each has a different unit of observation, different features, and different intended use.

Model A: Per-Attempt GBM (the strict one)

Model A is a gradient-boosting machine trained on individual pitch-level steal attempts (roughly 30,000–60,000 rows across 2018–2026). Each row is one specific attempt: this runner, this pitcher, this catcher, this count, this inning. The model tries to predict whether that single attempt succeeded. To prevent data leakage — where the same runner's career-aggregate lead numbers appear in both training and test rows — the model uses 'group cross-validation by runner ID,' meaning a runner's attempts are never split across training and testing. This is the strictest, most honest design. Its AUC is around 0.57–0.59, which looks low. But consider: the baseline success rate in the data is already about 54–58% (managers only send runners on attempts they believe have a good chance), which means a coin flip already gets you 54%. The per-attempt model at 0.59 is barely above that. Individual SB attempts are highly noisy — a bad hop, a perfect slide, a pitcher glancing at the wrong moment — and these are not captured in any data source we have.

Model B: Season-Level GBM (the better performer)

Model B aggregates all of a runner's per-attempt features up to the season level and then asks: did this runner have a ABOVE-average success rate this season? One row per runner-season. A runner who attempted 25 steals and succeeded 22 times (88%) is a 'yes'; one who attempted 20 and succeeded 13 (65%) is a 'no.' By smoothing over a full season of attempts, the noise largely cancels out, and the real signal — is this person actually good at stealing — becomes clearer. Model B's de-leaked AUC is about 0.59 (full) and 0.61 (pre-2023). Earlier versions reported 0.66–0.70, but those runs carried duplicate runner-season rows that leaked across CV folds; v7 averages them away, so the lower number is the honest one. This is the model we report as the headline.

Model C: Plain-English Weights (the GLM)

Model C (Simple Logistic Regression) is not designed to be the best predictor. It is designed to be the most interpretable. Unlike the GBMs (which learn thousands of nonlinear interactions between variables), the GLM has one coefficient per feature, and those coefficients translate directly into the 'SB% Boost per Tier' and 'Odds Multiplier' columns you see in the model output. If you want to know whether catcher pop time matters more than jump time, read the GLM weights — the GBM will give a

The Three Models — A, B, and C

better prediction but won't tell you which feature drove the prediction.

AUC: Why ~0.59 (De-Leaked) Is Honest, Not a Regression

What AUC Measures, and the v7 De-Leaking

AUC (Area Under the ROC Curve) measures how well a classifier separates 'yes' from 'no'. 0.50 is a coin flip; 1.00 is perfect. v7 Model B scores about 0.59 on the full 2015–2026 dataset and 0.61 pre-2023. Earlier versions reported 0.66–0.70, but those runs carried duplicate runner-season rows — repeated Statcast split measurements for one player-season — that leaked across cross-validation folds and inflated the score. v7 averages those duplicates into one row per runner-season. The lower number is not a regression; it is the first honest estimate.

Why the Ceiling Is Real, Not a Bug

Stolen-base success has a large irreducible noise component. The throw, the catcher's grip on a given day, whether the pitcher noticed the runner leaning, the umpire's safe/out call on a bang-bang play — none of these appear in any public data feed. With public data the ceiling sits around 0.70–0.75; pushing higher would require private TrackMan or Hawk-Eye delivery-timing data that teams have internally but do not publish. The pre-2023 era scores a touch higher because the 2023 rule changes (larger bases, pickoff limits) shifted success rates league-wide and the model had to adapt to a new regime.

Known Limitations and Data Quirks

Lead Data Is Not Per-Season

The Savant base-stealing run value table is supposed to support year-by-year queries, but the year filter is silently ignored by the API — every query returns the same career snapshot regardless of what year you request. This means 'lead distance' and 'lead gain' in this model are career averages, not the specific numbers from a given season. For most runners this barely matters (their approach is consistent), but for a runner who dramatically changed their lead strategy mid-career, the per-season comparison is slightly misleading.

The 3-2 Count Artifact

Statcast's pitch-level data records the balls-strikes count at each pitch. Stolen base attempts, however, are not tagged as individual pitch rows — they are embedded in the 'description text' (des field) of the plate appearance's final pitch. As a result, every parsed SB attempt inherits the count from the LAST pitch of the at-bat, not the actual pitch the runner broke on. The 3-2 count is the most common last-pitch count, so 3-2 shows up as having a 1.9% attempt rate while every other count shows near-zero. This is a data artifact, not a real strategic pattern. The '3-2 Count Attempt Share' variable was retained in v6 as a rough proxy for late-count aggressiveness but should not be interpreted as a precise tactical measure.

Pitcher Delivery Time Unavailable

Per-pitcher delivery time from first movement to release is the single most valuable missing variable. Longer deliveries give runners more time to build speed before the catcher receives the ball. This data exists inside MLB's proprietary TrackMan system but is not published. Savant's 'pitch tempo' CSV was tested but found to have a bug: the median_seconds_onbase column equals median_seconds_empty for every pitcher, suggesting the tempo values do not actually condition on base state. The model uses a league-constant delivery proxy (1.30 seconds) instead. Any future version that gains access to real per-pitcher delivery times would likely push AUC significantly above the current 0.70 ceiling.